

Skills Summary

Solving Linear Systems by Elimination

Use this reference while completing your worksheet or when studying.

Skill 1 — Scaling one equation to match a pair

Rule: Multiplying *both sides* of an equation by a nonzero number gives an equivalent equation — same solutions. So you may rescale one equation until a pair of like terms are opposites, then add.

When one multiplication is enough: the coefficient of the chosen variable in one equation divides the one in the other. Multiply by the quotient, and match the signs so the pair cancels.

Example: Solve $2x + 5y = 19$ and $3x - y = 3$.

Step 1. The y -coefficients are 5 and -1 , and 1 divides 5 — so one multiplication is enough: scale the second equation, not both.

Step 2. Multiply the second equation by 5: $15x - 5y = 15$. Add it to the first: $17x = 34$, so $x = 2$.

Step 3. Substitute back into $3x - y = 3$: $6 - y = 3$, so $y = 3$. Check the first equation: $2(2) + 5(3) = 19$ ✓

$$x = 2 \quad \text{and} \quad y = 3$$

Try it: Solve $4x + y = 14$ and $3x - 2y = 5$.

check: $x = 3, y = 2$

Skill 2 — Scaling both equations with an LCM

Rule: When neither coefficient of the chosen variable divides the other, scale *both* equations to their least common multiple. Pick the variable whose LCM is smallest — that keeps the arithmetic light.

Signs: if the scaled terms end up with the *same* sign, subtract; if they end up *opposite*, add.

Example: Solve $3x + 2y = 16$ and $5x + 3y = 26$.

Step 1. Neither 3 nor 5 divides the other, so no single multiplication works — go to the LCM of the x -coefficients, which is 15.

Step 2. Multiply the first equation by 5 and the second by 3: $15x + 10y = 80$ and $15x + 9y = 78$. Same sign, so subtract: $y = 2$.

Step 3. Substitute into $3x + 2y = 16$: $3x + 4 = 16$, so $x = 4$. Check: $5(4) + 3(2) = 26$ ✓

$$x = 4 \quad \text{and} \quad y = 2$$

Try it: Solve $2x + 3y = 13$ and $5x + 4y = 22$.

check: $x = 1, y = 3$

(!) **Watch out:** Multiply *every* term, including the constant on the right. Scaling $3x + 2y = 16$ by 5 and writing $15x + 10y = 16$ is the single most common elimination error.

Skill 3 — Writing a system from a word problem

Model: name each unknown with its unit (“ a = price of one adult ticket, in dollars”), then write one equation per *purchase or trip* described. Two descriptions give two equations.

Read the answer back into the story — a negative price or a fractional person means an equation was mis-copied.

Example: Two adult and three child tickets cost \$34; four adult and one child ticket cost \$38. Find both prices in dollars.

Step 1. Two separate purchases of the same two items is a two-equation model — let a and c be the adult and child prices in dollars.

Step 2. $2a + 3c = 34$ and $4a + c = 38$. The c -coefficients are 3 and 1, so scale the second equation by 3: $12a + 3c = 114$; subtract the first: $10a = 80$, so $a = 8$.

Step 3. Then $4(8) + c = 38$ gives $c = 6$. Check the first purchase: $2(8) + 3(6) = 34$ ✓
 $a = 8$ dollars and $c = 6$ dollars

Try it: Three notebooks and two pens cost \$16; five notebooks and four pens cost \$28. Find both prices in dollars.

check: $3n = 16$, $5n = 28$